Education

2017 Dec. **B.S., Mathematics**, *University of Texas*, Austin, United States.

2017 Dec. **Certificate, Scientific Computation**, *University of Texas*, Austin, United States.

Experience

Research

2021 Aug. – Present **Engineering scientist associate**, *Texas Advanced Computing Center*, Austin, United States.
- Principal developer of the highest quality terrain data in Texas
- Acquired the thumbs-up for this data from top State hydrologists and eminent UT engineers
- Vigorously advanced University tech commercialization of this data, reasonably valued at $2M/customer
- Extended our reach with novel advertising that has activated dozens of new research relations across the University and private sector
- Achieved a 14% click-through rate (CTR) with key audiences
- Innovated an open-source licensing strategy for this data that is novel to the engineering community and protects University data producers from liability due to death or injury
- Cemented TACC's relationship with the US Army Corps of Engineers through key scaling results
- Teach undergrads thru post-docs from UT Austin, University of Cambridge, & Texas A&M - Galveston

2019 July – 2019 Aug. **Research intern**, *Information Sciences Institute, University of Southern California*, Marina del Rey, United States.
- Advanced collaboration on DARPA-funded Model Integration (MINT) project
- Supported implementation of surface and subsurface hydro models in Texas and Ethiopia

2018 Feb. – 2021 July **Research engineering / scientist associate I**, *Texas Advanced Computing Center*, Austin, United States.
- Contributed research in integrated modelling and water resource management in Texas and abroad
- Collaborated to unify data of Texas's water resources for integrated hydrology modellers

Teaching

2020 Aug. – 2020 Dec. **Co-instructor**, *University of Texas*, Austin, United States.
Scientific computation Consulted with students for their homework and class projects

2018 June – 2018 Aug. **Teaching assistant**, *Texas Advanced Computing Center*, Austin, United States.
Machine learning for the geosciences Recommended high performance optimisations for Petrobras's code

2018 Feb. – 2018 May **Co-instructor**, *University of Texas*, Austin, United States.
Data informatics and intelligent systems for the geosciences Designed and instructed the technical half of this graduate course: intermediate programming, Linux, and introductory high-performance computing

Peer-reviewed articles

2021 July Gil, Y., D. Garijo, D. Khider, C. A. Knoblock, M. Osorio, H. Vargas, M. Pham, J. Pujara, B. Shbita, B. Vu, Y. Chiang, D. Feldman, Y. Lin, H. Song, V. Kumar, A. Khandelwal, M. Steinbach, K. Tayal, S. Xu, S. A. Pierce, L. Pearson, *D. Hardesty Lewis*, E. Deelman, R. F. da Silva, R. Mayani, A. R. Kemanian, Y. Shi, L. Leonard, S. .D. Peckham, M. Stoica, K. M. Cobourn, Z. Zhang, C. Duffy, L. Shu. "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". The ACM Transactions on Interactive Intelligent Systems 11(2):11.

Conference papers

Daniel Hardesty Lewis
1915 Santa Clara st – Austin, TX 78757 – United States

2019 Mar. Garijo, D., J. Pujara, B. Vu, D. Feldman, R. Mayani, K. Cobourn, C. Duffy, A. R. Kemanian, L. Shu, V. Kumar, A. Khandelwal, D. Khider, K. Tayal, S. D. Peckham, M. Stoica, A. Dabrowski, *D. Hardesty Lewis*, S. A. Pierce, V. Ratnakar, Y. Gil, E. Deelman, R. F. da Silva, C. Knoblock, Y. Chiang, M. Pham. "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". ACM 24th International Conference on Intelligent User Interfaces (IUI'19).

2018 June Garijo, D., D. Khider, Y. Gil, L. A. M. C. Carvalho, B. T. Essawy, S. A. Pierce, *D. Hardesty Lewis*, V. Ratnakar, S. D. Peckham, C. Duffy, J. L. Goodall. "A Semantic Model Catalog to Support Comparison and Reuse". 9th International Congress on Environmental Modelling and Software.

Academic projects

2022 June – Present — Advanced compound flood hazard assessment science
Partner with US Army Corps of Engineers to statistically model flood hazards in Texas's coastal region

2021 Oct. – Present — Texas Disaster Information System (TDIS)
Provide analytical web-based spatial data system to support more resilient decision-making at the state level

2020 Nov. – Present — Networks for Hazard Preparedness and Response
Overlay new, finely tuned flood maps over specially designed building and social vulnerability maps

2020 Sep. – Present — Real-time flood inundation mapping to improve community resilience
Provide computationally efficient, high-resolution flood inundation maps to emergency response personnel

2017 Dec. – Present — MINT: Model INTegration Through Knowledge-Rich Data and Process Composition
Integrate geoscience models with models from other fields including agriculture, economics, and social sciences

Professional projects

2019 Nov. – Present — **Board member**, *Austin Community Reinvestment Cooperative (ACRC)*, Austin, United States.
Helped establish cooperative real estate investment firm

2019 May – Present — **Member-owner**, *Hardy Cooperative*, Austin, United States.
Helped establish 14-unit Central Austin affordable housing cooperative

2019 May – 2020 Apr. — **Member, housing committee**, *Austin Cooperative Business Association*, Austin, United States.
Austin Land Development Code (LDC) Revision

Students mentored

2021 Aug. – Present — Khushboo Agarwal
MS Energy and Earth Resources, the University of Texas at Austin (expected 2023)

2020 Aug. – Present — Mark Wang
PhD Civil Engineering, the University of Texas at Austin (expected 2025)
MS Environmental and Water Resources Engineering, the University of Texas at Austin (2022)

2019 Aug. – 2021 May — Robert Schomp
MS Environmental and Water Resources Engineering, the University of Texas at Austin (2021)

2019 Aug. – 2020 May — Alec Carruthers
MS Environmental Engineering Technology, the University of Texas at Austin (2020)

Technology skills

Languages — Fortran, Perl, Python, Shell, PL/pgsql

OSs — GNU/Linux, macOS, Windows

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Virtualization	Docker, Singularity, chroot
Scientific models	MODFLOW, HAND, HEC-RAS
Cloud	AWS, Azure
Databases	PostgreSQL
Libraries	GDAL/OGR, NumPy, GeoPandas, RasterIO, Matplotlib, Plotly, Dash
SCM	Git, SVN

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